

PRE BOARD EXAM (2024-25) MARKING SCHEME
CLASS- X
SUBJECT- ARTIFICIAL INTELLIGENCE (417)

Max. Time: 2 Hours
50

Max. Marks:

SECTION A: OBJECTIVE TYPE QUESTIONS

Q. NO.	QUESTION	UNIT/ CHAPTER NO.	MARKS
Q.1.	Answer any 4 out of the given 6 questions on Employability Skills (1 x 4 = 4 marks)		
i.	b. Self-Motivation	Unit 2	1
ii.	a. Real Time	Unit 3	1
iii.	b. Both A and R are true but R is not the correct explanation of A.	Unit 4	1
iv.	d. Adaptability	Unit 2	1
v.	a.e-mail	Unit 1	1
vi.	b. Sustainability	Unit 5	1
Q. 2	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)		
i.	b. Both A and R are true but R is not the correct explanation of A.	Unit 1	1
ii.	c. AI Bias	Unit 1	1
iii.	a. Both A and R are correct	Unit 2	1
iv.	b) Graphical Representation	Unit 2	1
v.	d. Data Science	Unit 4	1
vi.	a. Stop word	Unit 6	1

Q. NO.	QUESTION	UNIT/ CHAPTER NO.	MARKS
Q.3.	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)		
i.	b. Internet of Things	Unit 1	1
ii.	c. Scatter plot	Unit 2	1
iii.	a. Genetics and Genomics	Unit 4	1
iv.	c. Object detection	Unit 5	1
v.	b. Script Chatbot	Unit 6	1
vi.	d. (i) and (iv)	Unit 7	1
Q. 4.	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)		
i.	a. Deep learning	Unit 1	1
ii.	c. Statement 1 is correct but Statement 2 is incorrect	Unit 7	1
iii.	c. Overfitting	Unit 4	1
iv.	a. 256	Unit 5	1
v.	a. Sentiment Analysis	Unit 6	1
vi.	b. Training result is very good and the test result is poor	Unit 7	1
Q. 5.	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)		
i.	c. Natural Language Processing	Unit 1	1
ii.	a. Precision	Unit 7	1
iii.	b. The intensity	Unit 5	1
iv.	(c) Virtual Assistants are not applications of NLP	Unit 6	1
v.	c. Term Frequency and Inverse Document Frequency	Unit 6	1
vi.	c. False Negative	Unit 7	1

SECTION B: SUBJECTIVE TYPE QUESTIONS

Q. NO.	QUESTION	UNIT/ CHAPTER NO.	MARKS
Answer any 3 out of the given 5 questions on Employability Skills in 20 – 30 words each (2 x 3 = 6 marks)			
Q.6.	Enumerate any two ways to overcome barriers to effective communication. ANS-Ways to Overcome Barriers to Effective Communication (Any Two) a) Use simple language b) Do not form assumptions on culture, religion or geography c) Try to communicate in person as much as possible d) Use visuals e) Take help of a translator to overcome differences in language f) Be respectful of other's opinions 2 marks for any two correct points	Unit1	2
Q.7.	In SMART goals, what does 'S' stand for? Explain. ANS-We can use SMART method to set goals. SMART stands for: • Specific: A specific and clear goal answers six questions. Who is involved in the goal? What do I want to do? Where do I start? When do I start and finish? Which means do I use? Why am I doing this? Not a specific goal: "I would learn to speak English." Specific goal: "I would learn to speak English fluently by joining coaching classes after my school every day, and in six months I will take part in the inter-school debate competition." 2 Marks for correct explanation.	Unit 2	2
Q.8.	Aisha is searching some contents related to sports in search engine but during searching help her about how to prevent her data while using the internet? ANS: To protect the data following measures should be take care: • Use a strong password in your account. • Install antivirus and firewall • Keep the information in encrypted format • Click only on secure websites during internet surfing. 1 Mark each for describing any two correct options.	Unit 3	2
Q.9.	How does an entrepreneur promotes economic prosperity of a country? ANS. Entrepreneurs can bring about drastic changes in the very structure of the economy. They stand beyond challenges and make huge profits in every economic system. They are an important source of economic development. They create jobs, wealth and capital in the country. They promote investment, increase production and bring competitiveness in business, reduce costs of products and raise the standard of living in society 2 Marks for correct answer.	Unit 4	2
Q.10.	"With increasing population and income, the consumption of goods is increasing day by day." Comment. ANS-This has led to increase in production and utilization of natural resources, which are required for producing goods. Society must thus change its development strategy to a new form where development will not destroy the environment. 2 Marks for correct answer.	Unit 5	2

Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)			
Q.11.	How do you understand whether a machine/application is AI based or not? Explain with an example. ANS: Any machine/application that is trained with data and can make decisions/predictions on its own can be termed as AI. Example: The bot or automation machine is not trained with any data is not an AI. while a chatbot that understands and processes human language is an AI. 1 mark for correct answer, 1 mark for correct example.	Unit 1	2
Q.12.	Gaurav want to make a model which will organize dataset comprising of 100 images of apples and 100 images of bananas. It compare the image with the trained data and according to the labels of trained images, it will identify the test image as an apple and banana. Which learning model should he use and why? ANS: It refers to the AI modelling where the relationship or patterns in data are defined by the developer. The machine follows the rules or instructions mentioned by the developer, and performs its task accordingly. This is known as Rule based model. The rules given to the machine in this example are the labels given to the machine for each image in the training dataset. 1 mark for identifying correct modal, 1 mark for the correct justification.	Unit 2	2
Q.13.	What are the different types of sources of data from where we can collect reliable and authentic datasets? ANS: There could be many ways and sources from where we can collect reliable and authentic datasets namely Surveys, Web scrapping, Sensors, Cameras, Observations, Research, Investigation, API etc. Sometimes Internet is also used to acquire data but the most important point to keep in mind is that the data should be taken from reliable and authentic websites only. Some reliable data sources are UN, Google scholar, Finance, CIA, Data gov etc. 2 Marks for correct answer.	Unit 4	2

Q.14.	Explain how pixel density affects the visual quality of an image. ANS: Pixel density, measured in pixels per inch (PPI), affects the sharpness and clarity of an image. Higher pixel density means more pixels are packed into a given area, resulting in a crisper and more detailed image, which is particularly important for high-quality displays and prints. 2 Marks for correct answer.	Unit 5	2
Q.15.	Identify the stop words in the given sentence: Pollution is the introduction of contaminants into the natural environment that cause adverse change. The three types of pollution are air pollution, water pollution and land pollution. Mail details pollu@gmail.com ANS-is,the,of,that,are,and,into ½ Mark each for describing any four stopwords.	Unit 6	2

Q.16.	<table><tr><th colspan="2" rowspan="2">CONFUSION MATRIX</th><th colspan="2">Reality</th></tr><tr><th>True</th><th>False</th></tr><tr><th rowspan="2">prediction</th><th>True</th><td>10</td><td>15</td></tr><tr><th>False</th><td>5</td><td>3</td></tr></table>	CONFUSION MATRIX		Reality		True	False	prediction	True	10	15	False	5	3	Unit 7	2
CONFUSION MATRIX				Reality												
		True	False													
prediction	True	10	15													
	False	5	3													
Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)																
Q.17.	Differentiate between what is AI and what is not AI with the help of an example? ANS- <table><tr><th>AI Machine</th><th>Not AI machine</th></tr><tr><td>1. AI machines are trained with data and algorithm. 2. AI machines learn from mistakes and experience. They try to improvise on their next iterations. 3. AI machines can analyse the situation and can take decisions accordingly. 4. AI based drones capture the real-time data during the flight, processes it in real-time, and makes a human-independent decision based on the processed data.</td><td>1. Smart machines which are not AI, do not require training data, they work on algorithms only. 2. Smart machines work on fixed algorithms and they always work with the same level of efficiency, which is programmed into them. 3. Machines which are not AI cannot take decisions on their own. 4. An automatic door in a shopping mall, seems to be AI-enabled, but it is built with only sensor technology.</td></tr></table> 1 Mark each for describing each difference.	AI Machine	Not AI machine	1. AI machines are trained with data and algorithm. 2. AI machines learn from mistakes and experience. They try to improvise on their next iterations. 3. AI machines can analyse the situation and can take decisions accordingly. 4. AI based drones capture the real-time data during the flight, processes it in real-time, and makes a human-independent decision based on the processed data.	1. Smart machines which are not AI, do not require training data, they work on algorithms only. 2. Smart machines work on fixed algorithms and they always work with the same level of efficiency, which is programmed into them. 3. Machines which are not AI cannot take decisions on their own. 4. An automatic door in a shopping mall, seems to be AI-enabled, but it is built with only sensor technology.	Unit 1	4									
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Q.18.	What is 4Ws Problem Canvas? Explain Problem Statement Template. ANS: Who, What, Where, and Why are the 4Ws of problem scoping. These Ws aid in more accurate and effective problem identification and comprehension. who : who is facing for problem who are the stakeholders of problem. what: what is refer to a asking question. where: where is refer to asking about the place where the person was going. why: why is refer to a asking about the person like why are you asking question. The Problem Statement Template helps us to summarize all the key points into one single template. So that in the future, whenever there is a need to look back at the basis of the problem, we can take a look at the Problem Statement Template and understand its key elements of it. ½ Mark each for describing each of the Ws. 2 marks for correct explanation of problem statement template.	Unit 2	4													

Q.19.	Identify and explain the types of the learning-based approaches in the given figures. ANS: Reinforcement Learning It's a learning by experiencing. This means that the machine carries out a task and then observes if it was a mistake or if it got a reward. So it's a Reward-Based learning model. Ex- When some sort of signal is provided to the algorithms that associates good behaviour with positive signal and bad behaviour with a negative one, the algo can	Unit 2	4
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	be reinforced to prefer good behavior over bad behaviour. Over some time, the algo learns and make fewer mistakes. 1 mark for correct identification and 3 marks for correct explanation.															
Q.20.	<table><tr><th>Human language</th><th>Computer Language</th></tr><tr><td>Human language is made up of letters, words and sentences depending on the languages.</td><td>Machine/computer understands the language of numbers (binary numbers- 0'sand 1's). Everything that is sent to the machine has to be converted to numbers.</td></tr><tr><td>It is very easy for humans to process andcommunicate in natural languages like English, Hindi etc.</td><td>For machines understanding and generatingnatural languages is very complex process.</td></tr><tr><td>Our brain keeps on processing the sounds that it hears around itself and tries to make sense out of them all the time.</td><td>Computer uses NLP techniques like Text Normalisation, Bag of words to convert the text to numbers for it to process.</td></tr></table>	Human language	Computer Language	Human language is made up of letters, words and sentences depending on the languages.	Machine/computer understands the language of numbers (binary numbers- 0'sand 1's). Everything that is sent to the machine has to be converted to numbers.	It is very easy for humans to process andcommunicate in natural languages like English, Hindi etc.	For machines understanding and generatingnatural languages is very complex process.	Our brain keeps on processing the sounds that it hears around itself and tries to make sense out of them all the time.	Computer uses NLP techniques like Text Normalisation, Bag of words to convert the text to numbers for it to process.	Unit 6	4					
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Q.21.	An IT company situated in Bombay developed an AI model which predicts the purchasing of electronic gadgets. During testing, the AI model came up with the following predictions. Based on the given predictions, calculate the following:- <table><tr><th colspan="2" rowspan="2">Confusion Matrix</th><th colspan="2">Reality</th></tr><tr><th>Yes</th><th>No</th></tr><tr><td rowspan="2">Prediction</td><td>Yes</td><td>60</td><td>25</td></tr><tr><td>No</td><td>5</td><td>10</td></tr></table> Precision = True Positives / (True Positives + False Positives) = 60 / (60 + 5) = 0.923 Recall = True Positives / (True Positives + False Negatives) = 60 /(60 + 25) = 0.706 F1 Score = 2 * (Precision * Recall) / (Precision + Recall) = 2 * (0.923 *0.706) / (0.923 + 0.706) ≈ 0.801 (ii) Total tests performed = Sum of all entries in the confusion matrix = 60 +25 + 5 + 10 = 100 1 mark each for each correct evaluation of precision, recall, F1score, Total test performed.	Confusion Matrix		Reality		Yes	No	Prediction	Yes	60	25	No	5	10	Unit 7	4
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